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# Assignment - 01

## Course : CSE260

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## Ans to ques - 1

(a) Binary to decimal conversion,

$(1011100100001111)_2$

$= 1\times2^0+1\times2^1+1\times2^2+1\times2^3+0\times2^4+0\times2^5+0\times2^6+1\times2^7+0\times2^8+0\times2^9+$
$1\times2^{11}+1\times2^{12}+1\times2^{13}+0\times2^{14}+1\times2^{15}$

$= 1+2+4+8+0+0+0+128+0+0+1024+2048+4096+$
$0+16384$

$=(23695)_{10}$

(b) $(11011.10101)_2$ to decimal conversion.

Now, $(11011)_2 = 1+2+0+8+\cancel{32}\,16 = (27)_{10}$

fraction,

$.10101 = 1\times2^{-1}+0\times2^{-2}+1\times2^{-3}+0\times2^{-4}+1\times2^{-5}$

$= 65625$

So, $(11011.10101)_2 = (27.65625)_{10}$

## Ans to ques-2

2) Here, $(8195)_{10} =$

$$
\begin{array}{r|l}
2 & 8195 \\
2 & 4097 \\
2 & 2048 \\
2 & 1024 \\
2 & 512 \\
2 & 256 \\
2 & 128 \\
2 & 64 \\
2 & 32 \\
2 & 16 \\
2 & 8 \\
2 & 4 \\
2 & 2 \\
 & 1
\end{array}
\quad 1100000000000001 \leftarrow msb
$$

So, $(8195)_{10} = (10000000000011)_2$ (ans)

## Ans to ques-3

3) $(9126)_{10}$ to hexa decimal,

$$
\begin{array}{r|l}
16 & 9126 \\
16 & 570 \\
16 & 35 \\
 & 2
\end{array}
\quad 6A32 \leftarrow msb
$$

So, $(9126)_{10} = (23A6)_{16}$ (ans):

## Ans to ques - 04

# Base conversion.

(a) $(29)_{12} = (?)_8$

Now, $(29)_{12} = 2\times12^1 + 9\times12^0$

$= (33)_{10}$

and Now,

$$8 \,\underline{|\, 33 \,|}\, 1\ 4 \leftarrow msb$$
$$4$$

$= (41)_8$

$\therefore (29)_{12} = (41)_8$ ans):

(b) $(102101110201)_5 = (?)_4$

Now, $(102101110201)_5 = 1\times5^{11} + 0\times5^{10} + 2\times5^9 + 1\times5^8 + 0\times5^7 + 1\times5^6 + 1\times5^5 + 1\times5^4 + 0\times5^3 + 2\times5^2 + 0\times5^1 + 1\times5^0$

$= (51191276)_{10}$

Now,

$$\begin{array}{r|l|l}
4 & 51191276 & 0323131013003 \leftarrow msb \\
4 & 12797819 & \\
4 & 3199454 & \\
4 & 799863 & \\
4 & 199965 & \\
4 & 49991 & \\
4 & 12497 & \\
4 & 3124 & \\
4 & 781 & \\
4 & 195 & \\
4 & 48 & \\
4 & 12 & \\
 & 3 &
\end{array}$$

$= (3003101313230)_4$ ans.

so.

## Ans to ques - 5

Here, $(812)_{17}$, $(138)_{17}$, and, $(812)_{17} = (2331)_{10}$
$(138)_{17} = (348)_{10}$

**Addition:**

$$\begin{array}{r} (812)_{17} \\ +(138)_{17} \\ \hline (94A)_{17} \end{array}$$

$= (2679)_{10}$

and = $(2331)_{10}$

$$\begin{array}{r} (2331)_{10} \\ +(348)_{10} \\ \hline (2679)_{10} \end{array}$$

answer = $(94A)_{17}$

$10/17 = 0$
$10\%17 = 10 = A$

# Substraction,

$$\begin{array}{r} (812)_{17} \\ -(138)_{17} \\ \hline (6EB)_{17} \end{array}$$

$= (1983)_{10}$

$$\begin{array}{r} (2331)_{10} \\ -(348)_{10} \\ \hline (1983)_{10} \end{array}$$

$\therefore$ Answer = $(6EB)_{17}$

$(17+2-8)/17 = 0$
$(17+2-8)\%17 = 11 = B$
$(17+0-3)\%17 = 14 = E$
$(17+0-3)/17 = 0$

# Multiplication:

$$\begin{array}{r} (812)_{17} \\ \times(138)_{17} \\ \hline 3D8G \\ 1736\times \\ 812\times\times \\ \hline (9C1EG)_{17} \end{array}$$

$= (811288)_{10}$

$$\begin{array}{r} (2331)_{10} \\ \times(348)_{10} \\ \hline (811188)_{10} \end{array}$$

$16/17 = 0$
$16\%17 = 16 = G$
$64\%17 = 13 = D$
$64/17 = 3$

$14/17 = 0$
$14\%17 = 14 = E$
$(13+3+2)/17 = 1$
$(13+2+3)\%17 = 1$
$12\%17 = 12$
$12/17 = 0$

Answer = $(9C1EG)_{17}$

## Ans to ques-06

Here, if we substract 29 from 17 the equation will be, $(17)_{10} - (29)_{10} = (17)_{10} + (-29)_{10}$

Here, 17 is a positive signed number, but $(29)_{10}$ is negative signed number.

Now,

$(17)_{10} = (10001)_2$

$= (000010001)_2$ [9 bits

$= (000010001)_{1's\ comp}$ [9 bit 1's complement.

$(-29) = -(11101)_2$

$= -(000011101)_2$ [in 9 bits

$= (111100010)_{1's\ com}$ [9 bits 1's complement)

$= (111100011)_{2's\ com}$ [9 bits 2's complement of -29]

Now, $(17)_{10} = 000010001$

$(-29)_{10} = 111100011$

$(-12)_{10} = (111110100)_{2's\ comp}$

⊛ There has been no over flow as we added a positive (17) and negetive (-29) numbers.

⊛ The number of bit has not changed so there has been no over flow.

## Ans to ques no - 7

7) Here, the given numbers are both positive integers. so, if we add them, the equation will be = 15+28, Now,

$(15)_{10} = (1111)_2$

$= (001111)_2$ [in 6 bits]

$= (001111)_{1's\ com}$ [in 6 bits 1's complement]

$(28)_{10} = (11100)_2$

$= (011100)_2$ [in 6 bits]

$= (011100)_{1's\ com}$ [in 6 bits 1's compliment]

Now,

$$\begin{array}{r} (28)_{10} \\ +(15)_{10} \\ \hline (43)_{10} \end{array}, \quad = \begin{array}{r} (001111)_{1's} \\ +(011100)_{1's} \\ \hline (101011)_{1's} \end{array}$$

$= -(010100)_2$

$= -(20)_{10}$

"SO, there has been an Overflow in this case."

* We added two positive numbers but we got a negative number as sum. So there has been an overflow here.

* The bit number has also increased.

Ans to ques no- 08

@ 91-499,

here 499 is a negative signed number.

Now, in 1's complement,

$(91)_{10} = (1011011)_2$

$= (00001011011)_2$ [in 11 bit's]

$= (00001011011)_{1's\ com}$

Now,

$-(499) = -(111110011)_2$

$= -(00111110011)_2$

$= (11000001100)_{1's\ com}$

$= (11000001101)_{2's\ com}$

In 1's complement = $(00001011011)_{1's\ com}$

$(11000001100)_{1's\ com}$

$(11001100111)_{1's\ com}$

in 2's complement,

$(00001011011)_{2's\ com}$

$(11000001101)_{2's\ com}$

$(11001101000)_{2's\ com}$.

There has been no overflow in both cases here, becase the operation were done between a positive and negative signed number's.

~~Ans to ques no 8~~

8) b) 379+98,

Here ~~the~~ both the number are positive, hence 1's and 2's complement will give the same result.

So, $(379)_{10} = (101111011)_2$

$= (00101111011)_{1's\ com}$ [in 11 bit 1's complement]

and,

$(98)_{10} = (1100010)_2$

$= (00001100010)_2$ [in 11 bit]

$= (00001100010)_{1's\ com}$ [in 1's complement]

Now,

$$
\begin{array}{r}
(00101111011) \\
(00001100010) \\
\hline
(00111011101)_{1's\ com.}
\end{array}
$$

The addition of these two number gave us a positive ~~numb~~ signed number. So, there has been no overflow in the case.

Ans to ques-09

2) The cost of each Ram = $(1C2)_{16}$

$= (450)_{10}$

The cost of two Rams = $(450)_{10} \times 2$

$= (900)_{10}$

Cost of Graphics Card = $(10010110000)_2$

$= (1200)_{10}$

total cost = $(900)_{10}$

$+ (1200)_{10}$

$(2100)_{10}$

Money given by friend = $(4064)_8$

$= (2100)_{10}$

~~Aton~~ Remaining money = $(2100)_{10} - (2100)_{10}$

$= 0$

(ans:)